

Alejandro Sánchez Yalí

AI Engineer



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SUMMARY

AI Engineer and mathematician with 8+ years building software and 3+ years shipping LLM systems end to end. Lead engineer of Aluna, an AI recruitment platform: conversational screening agents over WhatsApp, web chat and email, a durable analysis pipeline with checkpointed retries, and per-agent model routing. Works in TypeScript and Python, and owns the architecture as well as the code. M.Sc. in Mathematics and 11 years teaching machine learning.

TECHNICAL SKILLS

- **AI & LLMs:** LLM agents, LangChain, LangGraph, LiteLLM, RAG, MCP, Claude, Gemini, OpenAI API, Whisper, ElevenLabs, agentic development with Claude Code
- **ML & Mathematics:** PyTorch, TensorFlow, Scikit-learn, JAX, Flax, NumPy, Pandas, linear algebra, probability, statistical inference
- **Languages:** Python (8+ years), TypeScript (6+ years), JavaScript (6+ years), SQL
- **Backend:** FastAPI, Django, Flask, NestJS, Node.js, RESTful APIs, microservices, event-driven design
- **Data & Infrastructure:** PostgreSQL, Redis, MongoDB, Docker, AWS, GCP, GitHub Actions, CI/CD
- **Architecture:** Clean architecture, DDD, SOLID, modular monolith, screaming architecture

EXPERIENCE

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| AI Specialist
<i>Lapzo</i> | Aug 2025 – Present
<i>Remote, Mexico</i> |
| <ul style="list-style-type: none">– Designed and built a Digital Professor that holds real-time conversations with learners and explains course content through interactive slides– Owned the architecture connecting LLM services, ElevenLabs voice synthesis and the real-time communication layer– Built an AI-powered educational content generator with LangChain, LangGraph and n8n workflow automation | |
| Independent Software & AI Consultant
<i>Freelance</i> | Aug 2024 – Present
<i>Remote</i> |
| <ul style="list-style-type: none">– Lead engineer of Aluna (AI recruitment platform, see below) and builder of a multi-tenant AI customer service agent for CREARIA (Colombia) on WhatsApp Cloud API, LiteLLM, RAG and MCP with automatic human escalation– Led technical design sessions with clients, translating business requirements into system architectures and naming the trade-off behind each technology decision– Use Claude Code daily on live client projects, steering and critically reviewing generated output to ship end-to-end features without eroding architectural boundaries– Designed database architecture for a client's workflow automation platform and served as code reviewer for their development team | |
| Full Stack Engineer
<i>Monadical</i> | Apr 2021 – Aug 2024
<i>Remote, Canada/US</i> |
| <ul style="list-style-type: none">– Built and integrated AI services for information retrieval with large language models (OpenAI, LangChain, Whisper) in production web applications– Participated in system design and architectural decisions, evaluating trade-offs across performance, scalability and maintainability– Built and scaled microservices for AI/ML and data-intensive applications; designed ETL pipelines and custom Django ORM extensions– Led code reviews and mentored engineers on a distributed team across Canada, the US and Latin America | |
| Full Stack Engineer
<i>BCFort</i> | Aug 2018 – Oct 2020
<i>Medellín, Colombia</i> |
| <ul style="list-style-type: none">– Designed system architecture for blockchain platforms: smart contract patterns, data models and service boundaries | |

- Developed blockchain and analytics platforms with Python (Django), Node.js, React and Web3.js on AWS, GCP and IBM Cloud

Professor of Mathematics & Machine Learning

Universidad de Antioquia

Jan 2010 – Oct 2021

Medellín, Colombia

- Taught Machine Learning, Data Science and Mathematics with Python, NumPy, Pandas, TensorFlow, PyTorch and Scikit-learn
- Developed the AI/ML elective curriculum for the Mathematics Institute and mentored student research in ML and data analysis

FEATURED PROJECTS

- **Aluna – AI Recruitment Platform for Staffing Agencies**  – *Next.js 15, TypeScript, FastAPI, LiteLLM, PostgreSQL, Inngest, Claude, Gemini, WhatsApp Cloud API*
 - Lead engineer of an AI recruitment platform for staffing agencies, built in 3 months – 336 source files across 4 packages, 26 tables and 11 external providers – split into a Next.js 15 application owning all domain state and a stateless FastAPI service owning model calls and the WhatsApp channel
 - Durable CV analysis pipeline on Inngest: 9 checkpointed steps with 3 retries, early exits for revoked consent, and deduplication placed before quota enforcement so a re-analysed CV reuses its cached score instead of paying for another model call
 - One conversational screening engine serving three transports – WhatsApp, signed-link web chat and email – so the conversation logic has a single implementation regardless of channel
 - Routed each agent to a deliberately chosen model through LiteLLM (Gemini for document analysis, Claude Sonnet for conversation, Claude Haiku for the vacancy assistant), overridable per agent at runtime from the database
- **Plixiq – AI Customer Support on WhatsApp**  – *Python, FastAPI, LiteLLM, RAG, PostgreSQL, Redis, Next.js*
 - Sole architect and engineer of a multi-tenant SaaS where AI agents resolve routine customer inquiries in about 2 seconds and escalate the rest to human operators with full conversation context
 - Modular monolith with 10 DDD bounded contexts enforced by a lint rule in CI; multi-provider LLM layer (Groq default, OpenAI fallback) so switching providers is a config change
- **Digital Human – Interactive AI Avatar**  – *React, Three.js, React Three Fiber, OpenAI GPT, Whisper, ElevenLabs*
 - Open-source conversational avatar with 464 stars and 110 forks: real-time speech-to-text, generation, voice synthesis and lip synchronisation
 - The engineering problem was latency budgeting across four services, since a human notices any pause in a conversation

OPEN SOURCE

- **talking-avatar-with-ai**  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- **Monadical-SAS/Morpheus**  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion
- 493 GitHub stars across own open-source projects – top 4.8% of ranked GitHub profiles (GitRanks)

CONFERENCE TALKS

- **Scaling Learning Models: Parallelism Strategies in JAX with Flax**  – PyCon Colombia, 2024

TECHNICAL WRITING

- **Pre-training: the Idea Behind Every LLM** (2026)
- **Neural Networks as DAGs of Parameterized Computational Programs** (2024)
- **Revolutionizing Animation: Building Digital Humans with LLMs** (2024)

EDUCATION

M.Sc. in Mathematics

Universidad de Antioquia

Aug 2010 - Sep 2013

Medellín, Colombia

B.Sc. in Teaching Mathematics and Physics

Universidad de Antioquia

Jan 2004 - Oct 2009

Medellín, Colombia